Cell communication and cell signaling

Prof. Samar H kassem

Applied Health Sciences Technology

- A **signaling molecule** is a chemical that cells use to communicate with each other and coordinate biological processes. These molecules transmit information by binding to specific receptors on target cells, triggering a series of intracellular events that lead to a response.

plasma membrane receptor is a protein located on the cell surface that binds to extracellular signaling molecules (ligands) and initiates a cellular response. These receptors are crucial for communication between cells and their environment.

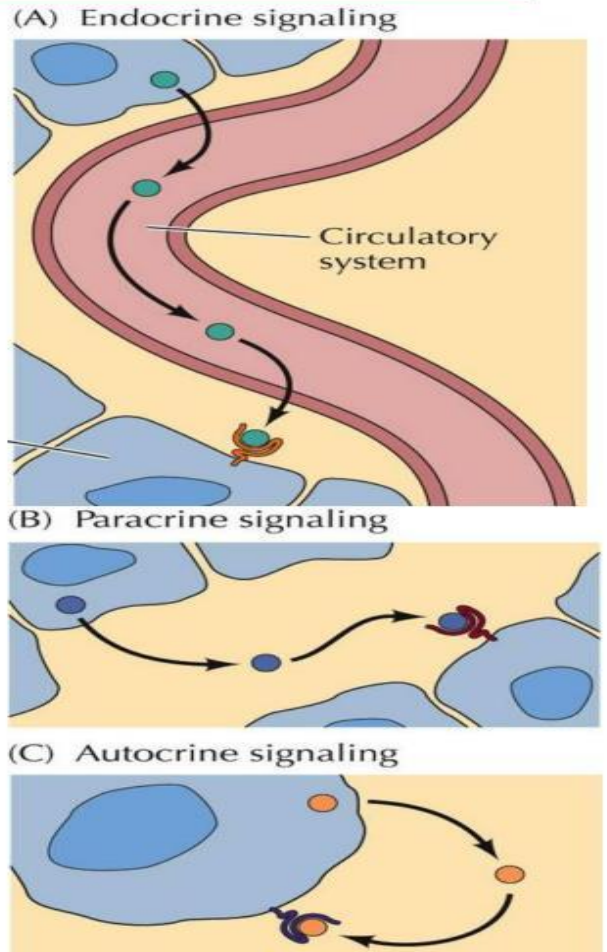
- _____

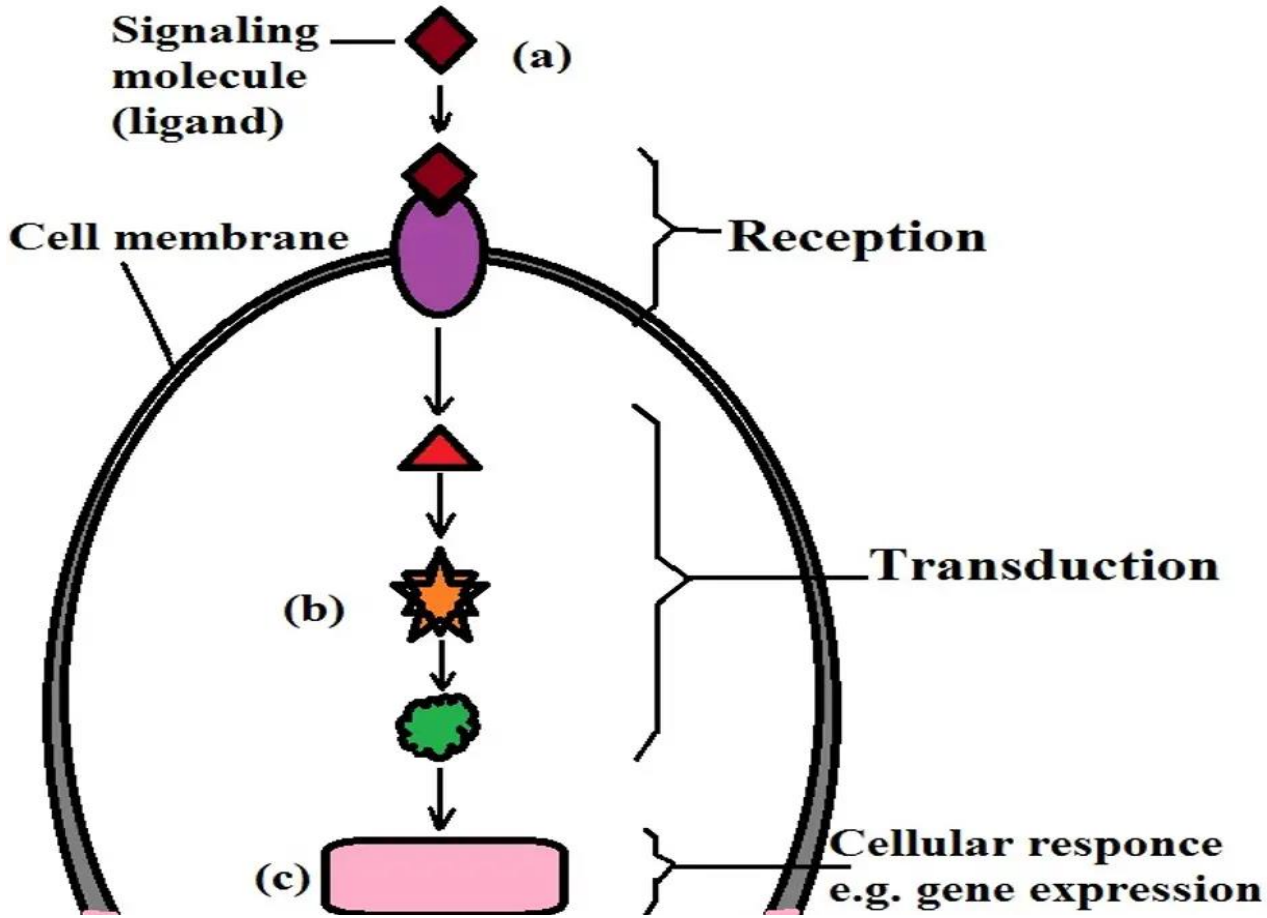
Types of Signaling Molecules

1. **Hormones** – Long-range messengers secreted by endocrine glands (e.g., insulin, adrenaline, thyroxin and cortisol).
2. **Neurotransmitters** – Short-range signals between nerve cells (e.g., dopamine, acetylcholine).
3. **Cytokines** – Immune system mediators (e.g., interleukins, interferons).
4. **Growth Factors** – Regulate cell growth and differentiation (e.g., PDGF, EGF).
5. **Second Messengers** – Intracellular signaling molecules (e.g., cAMP, Ca^{2+}).

Signaling by secreted molecules

- **Endocrine signaling**—signaling molecules (**hormones**), target cells at distant body sites Example: estrogen.
- **Paracrine signaling**—act on neighboring target cells, e.g., neurotransmitters
- **Autocrine signaling**—respond to self signaling molecules
  - T lymphocytes: proliferation in response to cytokines produced by the same cell



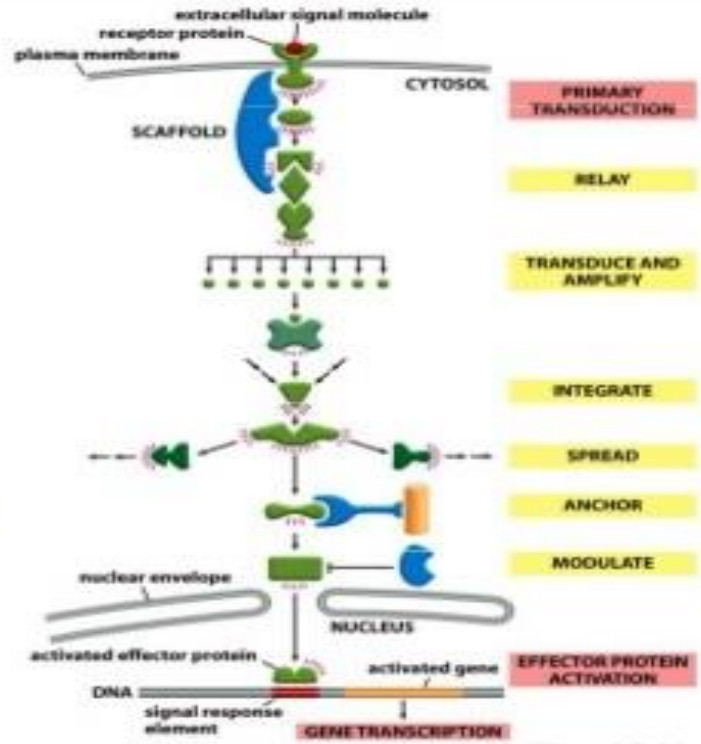

Signaling molecule (ligand)
(a)
Cell membrane
Reception
Transduction
(b)
(c)
Cellular responce
e.g. gene expression

Key Steps in Signal Transduction

1. **Signal Reception** – A signaling molecule (ligand) binds to a **specific receptor** on the cell membrane or inside the cell.
2. **Signal Transduction** – The binding event triggers a cascade of **biochemical changes**, often involving:
    1. Activation of **second messengers** (e.g., cAMP, Ca^{2+}, IP_3)
    2. Phosphorylation by **protein kinases** (e.g., MAPK, PKA, PKC)
    3. Conformational changes in **intracellular proteins**
3. **Cellular Response** – The signal leads to a specific outcome, such as:
    1. Gene expression changes
    2. Cytoskeletal reorganization
    3. Metabolic regulation
    4. Cell growth, differentiation, or apoptosis

Overview of Cell Signaling

- Cascade of signaling events
  - ↓Receptors
  - ↓Signal transducers
  - ↓Effector proteins
- Relay, integration, and distribution of signals require transducers.
- Signaling pathways regulate nearly all cellular functions.

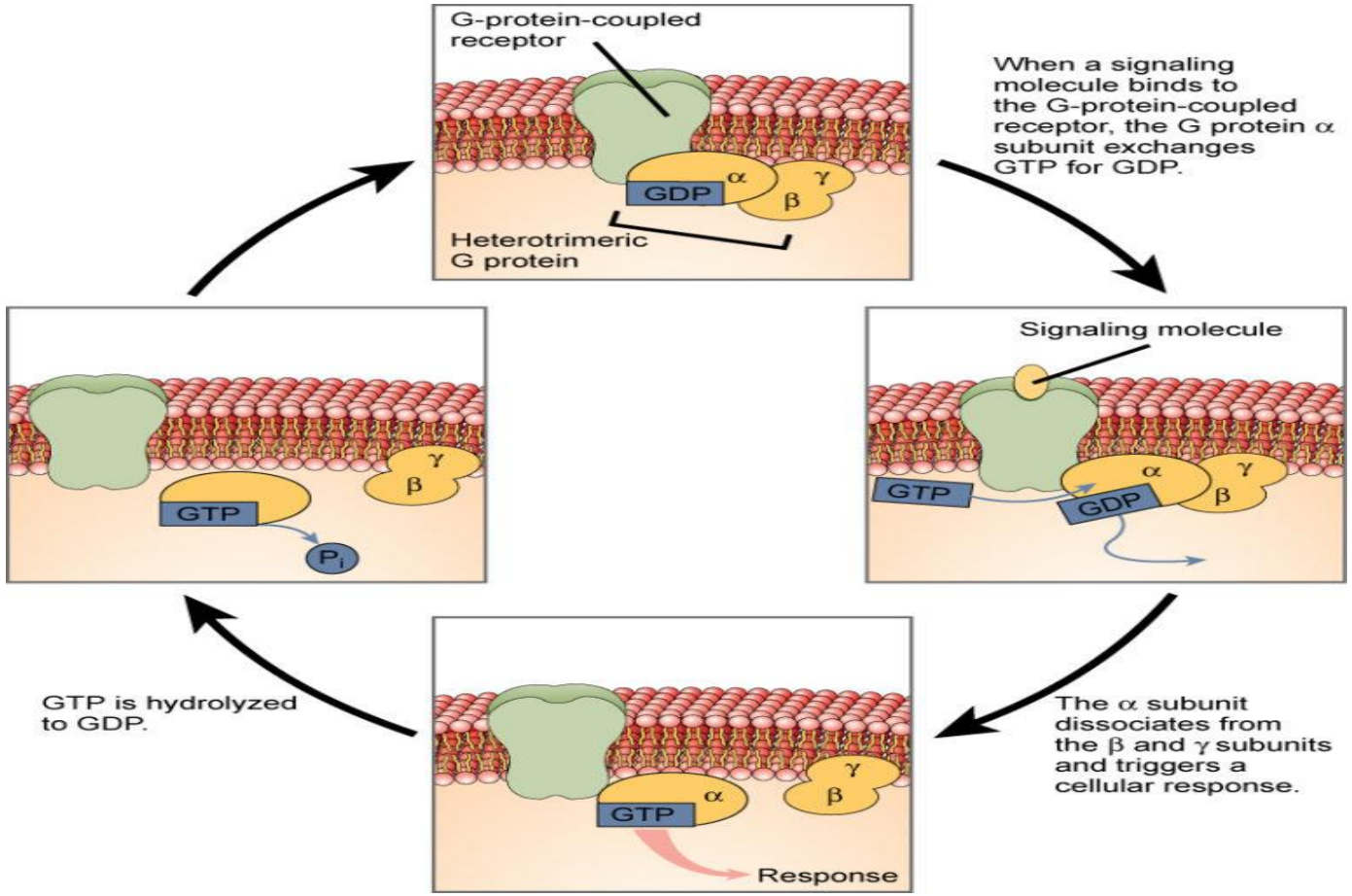


General Mechanism of Plasma Membrane Receptor Activation

1. **Ligand Binding** – The signaling molecule binds to the extracellular domain.
2. **Conformational Change** – The receptor changes shape to transmit the signal.
3. **Intracellular Signaling Cascade** – Activation of second messengers (e.g., cAMP, Ca^{2+}) or phosphorylation events.
4. **Cellular Response** – Can include gene expression changes, metabolic adjustments, or cytoskeletal reorganization.

Assignment **examples of PMRA**

G-protein signaling



G- protein signaling

